

Counting every degree of freedom of the plasma of `TransitionListener` exactly once

Notes on the degrees of freedom, the transverse photon and the scale factor

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1 What is counted where

TL keeps two inventories for the degrees of freedom in the plasma. The fields appearing in the effective potential are counted mode by mode, each polarization of a gauge boson separately, with their field- and temperature-dependent masses. Everything else is counted by the two tabulated radiation baths, the kinetically coupled bath `kin_coupled_*` (by default the Standard Model bath) and a kinetically decoupled bath `kin_decoupled_*` (which can be a secluded dark sector). Every particle which does not appear in the effective potential must appear in exactly one of the two. For a potential that contains Standard Model fields (for instance for the 2HDM models these are W , Z , t , b , τ , h and the Goldstone modes, which are flagged as `is_SM`) the same fields were counted twice in some places and not at all in others: the energy density counted the photon both in the potential and in the tabulated bath, while the entropy counted essentially none of them: the frozen tables still hold what is left of the τ and the b at the freezing temperature, but nothing of W , Z , t and h .

In the Landau gauge the modes appearing in the potential are not necessarily physical particles: each gauge boson contributes two transverse and one longitudinal mode, each broken generator adds a Goldstone mode, and each gauge boson comes with a ghost that removes one bosonic degree of freedom. For the Standard Model fields of the 2HDM at the zero-temperature vacuum,

$$\underbrace{1}_h + \underbrace{1+2}_{G^0, G^\pm} + \underbrace{2+1}_{Z_T, Z_L} + \underbrace{2+1}_{\gamma_T, \gamma_L} + \underbrace{4+2}_{W_T, W_L} - \underbrace{4}_{\text{ghosts}} = 12 = \underbrace{1}_h + \underbrace{3}_Z + \underbrace{6}_{W^\pm} + \underbrace{2}_\gamma, \quad (1)$$

so the gauge-dependent modes cancel and the physical count remains; with the fermions t , b and τ included, this sums to $12 + \frac{7}{8} \cdot 28 = 36.5$ degrees of freedom at temperatures far above their masses. That number is the bridge between the two inventories.

2 The three corrections

1. The modes of the potential. `h_eff_DS` and `g_eff_DS`, i.e. the entropy and energy degrees of freedom, were originally calculated by masking out every field flagged `is_SM`, skipping the Goldstone modes and keeping the ghosts. They now count every mode and subtract one ghost per gauge boson, which is the counting `radiationEnergyDensity` has always used. For the 2HDM the masked fields were counted nowhere: 68 entropy degrees of freedom at $T = 50 \text{ GeV}$ instead of 93. Subtracting the ghosts introduces a sign problem that the old counting did not have: the ghosts are massless in the Landau gauge, while the Goldstone modes they cancel are not, so once every mode of the potential is heavy compared with the temperature the modes die away and the ghost term does not, and the difference turns negative. A negative number of degrees of freedom has no meaning: when every mode is much heavier than the temperature, the sector has left the plasma and its entropy and energy are zero, not less than zero. The new count is therefore floored at zero, which is the correct limit.

2. The Standard Model fields in the tabulated bath. To leave room in the bath for the Standard Model fields that the potential already contains, TL froze the tabulated degrees of freedom above half the mass of the lightest massive `is_SM` field (0.888 GeV for the 2HDM), holding them at their value there for every higher temperature. Freezing is a blunt instrument for a selective job: it also froze the photon, the gluons, the light fermions and the QCD crossover, none of which the potential contains. The freezing temperature was also stored once for the whole module rather than per model, so in a session that builds several models the last one built decided which tables all of them used: constructing a dark model after a 2HDM removed the freezing again and silently changed the 2HDM’s degrees of freedom. It is replaced by subtracting exactly the Standard Model fields of the potential from the tables, at the masses they have in the zero-temperature vacuum. The inventories then close: in that vacuum the potential adds exactly what the table loses, so the total is the full Standard Model table plus the additional fields of the model. The subtraction depends only on the temperature once the model is built, so it is tabulated once per model and interpolated, to a relative accuracy of 3×10^{-7} .

3. The transverse photon of the 2HDM. The neutral gauge sector has two mass matrices in the (W^3, B) basis, not one. The longitudinal modes carry the Debye masses Π_{W_L} and Π_{B_L} , which are added to the matrix *before* it is diagonalised – since the two differ, the longitudinal mixing angle depends on the temperature and the longitudinal “photon” is not the photon direction. The transverse modes carry no thermal mass at this order, so their matrix is the same one without the Debye terms. TL took both eigenvalues of the longitudinal matrix, so the transverse photon carried the Debye mass of the longitudinal one. The transverse matrix has entries $g_2^2\phi^2/4$, $-g_1g_2\phi^2/4$ and $g_1^2\phi^2/4$; its determinant vanishes identically, so its eigenvalues are $(g_1^2 + g_2^2)\phi^2/4$ and 0 – the transverse Z , and a transverse photon that is massless at every field value, as electromagnetism being unbroken requires. The wrong mass entered the thermal potential and, with two degrees of freedom, the daisy term, so it moved the barrier of every 2HDM run.

3 The scale factor of the percolation integral

The percolation integral needs the scale factor $a(T)$ of the false vacuum, which TL obtained by integrating $d \ln a/dT = -1/(3c_s^2 T)$ with the trapezoidal rule over the solver’s temperature grid. That grid spans the overlap of the two traced phases, which for the 2HDM benchmark runs over thirteen decades with 71 points. Far below completion the sound speed of the false vacuum is meaningless – 43 of those 71 values were the silent fallback to $1/3$ – and the trapezoidal rule weights each interval by the value at its cold end, so one interval alone contributed 311 e-folds: $a(T)$ jumped by 10^{135} between neighbouring grid points and reached 10^{217} , and the integrator gave up with error 10, i.e. by an error marked (correctly) as stemming from a numerical source in the percolation solver.

The integral form of the same relation is entropy conservation, $a^3 s = \text{const}$, which needs no integration over the temperature grid, in order to obtain the scale factor directly: $a(T)/a(T_{\text{hot}}) = [s(T_{\text{hot}})/s(T)]^{1/3}$. The entropy is taken from the degrees of freedom of the transitioning sector rather than from $-\partial V/\partial T$. The reason is that the Arnold–Espinosa daisy term tends to $-T \sum_i n_i m_i^3/(12\pi)$ once the modes are heavy, a term linear in the temperature, whose contribution to the entropy $s = -\partial V/\partial T$ is therefore a constant. That constant cannot be physical: a species that is much heavier than the temperature is Boltzmann suppressed and carries an entropy that falls at least as fast as T^3 , so a plasma of such species cannot keep a temperature-independent entropy. It is an artefact of the high-temperature expansion the daisy term is derived from, and for the 2HDM benchmark it overstates the entropy of the plasma by a factor 3.7 at $T = 1$ GeV and by two orders of magnitude at $T = 0.1$ GeV.¹ Where the entropy cannot be used,

¹The artefact itself is left for the next pull request, which switches the Debye masses off as the particles that

the bag relation $a \propto 1/T$ continues the chain, and a step that would contract the universe as it cools is replaced by it as well. For a pure radiation bath, where $a \propto 1/T$ is exact, the old rule was wrong by a factor 680 at the cold end of a twelve-decade grid; the new one is exact.

4 What it changes

median change [%]	counting	photon	scale factor	all three
T_{crit}	0	-0.26	0	-0.26
T_{perc}	+0.02	-0.35	-0.001	-0.34
T_{reh}	+0.04	-0.27	-0.001	-0.27
α	+1.4	+1.4	+0.002	+2.9
$(\beta/H)_{R*}$	-1.0	-0.2	+0.03	-0.6
g_{eff} after reheating	-2.9	-0.05	-0.000	-2.9
h_{eff} after reheating	-1.8	-0.05	-0.000	-1.8
peak amplitude	+3.9	+3.7	-0.05	+7.1

Table 1: 2HDM type I line of the code comparison, 49 values of λ_3 , against the state before this pull request. The first column holds both halves of the counting, the modes of the potential and the subtraction from the tabulated bath, which were validated together; the second the transverse photon; the third the scale factor, which changes nothing but gives back the one point that the photon fix costs, so the line has 49 of 49 results.

In the dark sector models implemented in the code, the percolation is essentially untouched: along the abelian dark Higgs line T_{perc} , T_{reh} , α and β/H are bit-identical, and along the conformal line T_{perc} moves by $10^{-5}\%$ and T_{reh} by $+0.016\%$. What does move is the redshift: the degrees of freedom after reheating change by -0.9% (conformal) and $+0.3\%$ (abelian dark Higgs). Where the Goldstone modes are massless at the minimum the counting is unchanged to machine precision; where they are not, as in the conformal dark U(1) model, it changes by 16% at $T = 0.1 v$.

Still open. The daisy term is derived in the high-temperature expansion and leaves spurious temperature-independent pieces behind at $T \ll m$, which is why the sound speed of the broken phase drifts away from $1/3$ at low temperature instead of towards it. A following pull request will switch the Debye masses off as the particles that build them leave the plasma (see arXiv:2010.09744 and arXiv:2306.17239), validated on its own.

build them leave the plasma, following arXiv:2010.09744 and arXiv:2306.17239. Here it is only avoided, by not taking the entropy from the potential.